VEERA NAGA KOUSHIK CHALADI

AI / GENERATIVE AI ENGINEER
Agentic AI | RAG | LLM Applications | Python | AWS
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PROFESSIONAL SUMMARY

AI/GenAI Engineer with 5+ years of software engineering experience building enterprise-grade AI systems across agentic workflows, RAG, scientific evidence reasoning, decision intelligence, and scalable backend services. Strong hands-on experience with Python, FastAPI, LangGraph, LangChain, AWS, LLM orchestration, retrieval optimization, guardrails, and observability. Experienced in taking AI solutions from architecture and prototyping through production deployment, with a focus on grounded generation, reliability, performance, cost efficiency, and technical leadership across solution design and engineering delivery.

TECHNICAL SKILLS

Generative & Agentic AI: LLMs, Agentic AI, Multi-Agent Systems, Plan-and-Execute, ReAct, Agentic RAG, Tool Calling, Replanning, Human-in-the-Loop, Guardrails, MCP
Frameworks: LangChain, LangGraph
RAG & Retrieval: Hybrid Search, Semantic Search, Embeddings, Chunking, Metadata Filtering, Query Rewriting, Reranking, Context Optimization, Retrieval Evaluation
LLM Fine-Tuning - Working Knowledge: Hugging Face, SFT, PEFT, LoRA, QLoRA, Quantization
Predictive AI: Temporal Feature Engineering, Risk Scoring, Forecasting, Explainability, What-if Analysis
Backend & Data: Python, FastAPI, AsyncIO, REST APIs, Microservices, System Design, PostgreSQL, pgvector, DynamoDB, Vector Databases, OAuth2, AWS SNS
Cloud & LLMOps: AWS, EC2, ECS, EKS, RDS, Docker, Kubernetes, CI/CD, MLflow, OpenTelemetry, CloudWatch, Grafana

PROFESSIONAL EXPERIENCE

AMGEN | **Senior Associate Software Engineer**
Jun 2025 - Present

Scientific Results Reasoning Agent - Precision Medicine & Evidence Intelligence | Jun 2025 - Feb 2026

- Architected an agentic scientific-reasoning system that converts complex analysis outputs into evidence-grounded Precision Medicine insights and reviewer-ready briefs.
- Synthesized differential gene-expression, pathway-enrichment, immune-signature, correlation, and survival-style analyses across genes, pathways, immune signals, and clinical associations.
- Connected PubMed, public literature, internal knowledge, vector retrieval, and MCP tools to support traceable scientific interpretation.
- Cut LLM context and token usage by 20% through selective retrieval, evidence deduplication, reranking, and context optimization.
- Eliminated 10% of redundant LLM calls by reusing structured intermediate findings across retrieval, synthesis, and biological reasoning.
- Lowered end-to-end reasoning latency by 15% through parallel evidence retrieval, selective tool invocation, and context reuse.
- Mapped analytical observations to biological mechanisms, treatment-response patterns, resistance hypotheses, biomarker candidates, and mechanism-of-action context.
- Produced source-linked briefs with key findings, supporting evidence, biological interpretation, MoA context, and follow-up questions for SME review.

HRMAP AI - Enterprise Agentic AI & RAG Platform | Jun 2026 - Present

- Lead architecture and backend delivery for an enterprise agentic AI assistant that generates employee reviews from Workday data, feedback, recognitions, conversation history, and uploaded documents.

- Own implementation planning, design/code reviews, technical decisions, and engineering guidance for the AI/backend workstream.
- Orchestrate multi-step workflows for intent handling, retrieval, generation, revision, document Q&A, and grounded responses.
- Trim LLM context through metadata-aware retrieval, reranking, selective context injection, and query optimization.
- Accelerate responses with asynchronous execution, streaming, selective tool invocation, guardrails, and validation that prevent unnecessary model calls.
- Productionized Python/FastAPI services with persistence, failure handling, deployment, and observability.
- Cut AWS infrastructure spend by 29.4% and DynamoDB cost by an additional 10% through resource and architecture optimization.

TWFS Execution Readiness Command Center | Aug 2026 - Present

Side AI initiative | Hackathon prototype -> productized solution

- Co-developed a predictive execution-intelligence platform forecasting hiring, onboarding, proficiency, critical-mass, and capability go-live risks.
- Separated canonical data, temporal features, predictive analytics, decision logic, and LLM explanation into auditable architecture layers.
- Defined forecasts for hiring SLA breaches, onboarding dates, proficiency, critical-mass readiness, go-live risk, bottlenecks, and risk deterioration.
- Grounded explanations in measurable evidence, model contributions, business thresholds, reason codes, and recommended actions instead of free-form LLM reasoning.
- Constrained the agent layer to controlled evidence packets and analytics APIs for Q&A, executive summaries, and what-if analysis; helped productize the hackathon prototype.

PEGASYSTEMS | **Applications Engineer** **Feb 2023 - Jun 2025**

Pega AutoPilot - Agentic AI & Advanced RAG Platform

- Engineered an AI-driven automation platform using Python, FastAPI, Plan-and-Execute orchestration, multi-agent workflows, and RAG.
- Created a planner-executor workflow that decomposed requests, selected tools/retrieval operations, coordinated execution, and replanned on failure.
- Applied semantic/hybrid retrieval, embeddings, metadata filtering, query rewriting, reranking, and targeted context construction to improve relevance while limiting context-window usage.
- Modularized tools and retrievers to separate orchestration from application capabilities and reduce repeated prompt/context construction.
- Raised automation success rate by 20% through orchestration, retrieval, and tool-execution enhancements.
- Cut LLM API cost by 35% through token optimization, shorter context payloads, streamlined function execution, and fewer model calls.

PEGASYSTEMS | **Associate Applications Engineer** **Aug 2021 - Jan 2023**

Smart Claims Engine

- Built backend workflows, REST integrations, search, and data-processing components for enterprise claims automation.
- Implemented near-real-time indexing and optimized enterprise search and reporting workflows.

CERTIFICATIONS & AWARDS

Certifications: Google TensorFlow Developer Certification (2023) | Pega Certified Senior System Architect (2021)
Awards: Pega 4x4 Award Winner | Two-time Spot Award Winner

EDUCATION

B.Tech - Information Technology | Velagapudi Ramakrishna Siddhartha Engineering College (VRSEC) | GPA: 8.45/10 | 2017 - 2021